

SEQUENCE OF OPERATIONS AND FUNCTIONS OF TWO PUSHBUTTONS

Previously displayed on LCD	Button pressed	Displayed on LCD after pressing button	Action performed on relay board
Welcome! Press N	Next	Load: L1 Stat: OFF	--
Load: L1 Stat: OFF	Set	Load: L1 Stat: ON	Load 1 turned on
Load: L1 Stat: ON	Set	Load: L1 Stat: OFF	Load 1 turned off
Load: L1 Stat: OFF	Next	Load: L2 Stat: OFF	--
Load: L2 Stat: OFF	Set	Load: L2 Stat: ON	Load 2 turned on
Load: L2 Stat: ON	Set	Load: L2 Stat: OFF	Load 2 turned off
Load: L2 Stat: OFF	Next	Load: L3 Stat: OFF	--
Load: L3 Stat: OFF	Set	Load: L3 Stat: ON	Load 3 turned on
Load: L3 Stat: ON	Set	Load: L3 Stat: OFF	Load 3 turned off
Load: L3 Stat: OFF	Next	Load: L4 Stat: OFF	--
Load: L4 Stat: OFF	Set	Load: L4 Stat: ON	Load 4 turned on
Load: L4 Stat: ON	Set	Load: L4 Stat: OFF	Load 4 turned off

NC and C terminals are short-circuited, and NO and C terminals are open-circuited. When relay coil is energised, NC and C terminals are open-circuited, and NO and C terminals are short-circuited.

LCD. Arduino program uses the LCD in 4-bit mode. LCD data lines D4 to D7 are configured for reading data from Arduino. Pins 1 and 2 of the LCD are connected to Gnd and 5V, respectively, of Arduino. Read/write pin (pin 5) of the LCD is connected to Gnd. A 10kΩ preset is provided for adjusting the contrast of the display.

Software

Source code `lcd_interactive.ino` written in Arduino uses `LiquidCrystal.h` header file provided by Arduino library for working with the LCD.

`lcd.begin(16, 2)` function helps configure the 16 × 2 character LCD.

`attachInterrupt(0, nbpress, rising)` function calls interrupt handler `nbpress` whenever a signal connected to Interrupt 0 pin (pin 2) of Arduino makes a low-to-high, that is, rising-edge transition.

`attachInterrupt(1, sbpress, rising)`

EFY Note

The source code of this project is available for free download at source.efymag.com

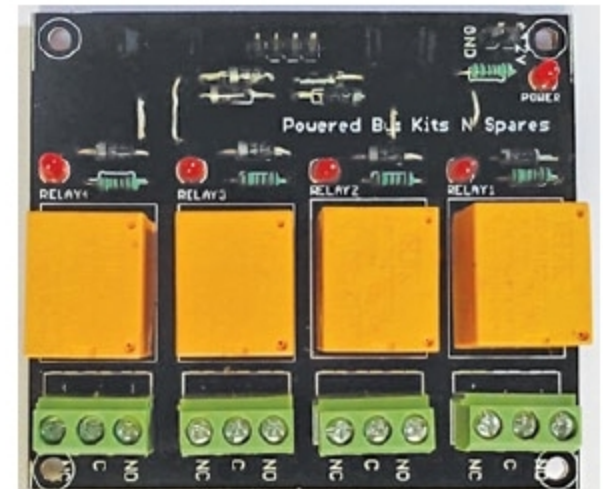


Fig. 3: Relay board

function calls interrupt handler `sbpress` whenever a signal connected to Interrupt 1 pin (pin 3) of Arduino makes a low-to-high, that is, rising-edge transition.

Construction and testing

Connect the circuit as shown in the circuit diagram. Compile and upload the program to Arduino from Arduino IDE. To power Arduino, use either a normal USB cable from USB port of the computer or an external 5V DC power supply source. Alternatively, use a 9V battery at Vin pin of Arduino to power it.

Connect the relay board at connector CON1 to connect the four inputs (IN1 through IN4) of the relay board to Arduino to control four electrical loads (L1 through L4). Note that, relay board used here requires a separate 12V DC power supply. **EFY**



Saikat Patra is passionate about electronics and MCU-based embedded system applications



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